

1.DATABASE SETUP AND IMPORT:

[3]

```
!pip install pymysql
```

```
Collecting pymysql
```

```
  Downloading PyMySQL-1.1.1-py3-none-any.whl.metadata (4.4 kB)
```

```
Downloading PyMySQL-1.1.1-py3-none-any.whl (44 kB)
```

```
----- 0.0/45.0 kB ? eta -:--:--
```

```
----- 20.5/45.0 kB ? eta -:--:--
```

```
----- 45.0/45.0 kB 739.0 kB/s eta 0:00:00
```

```
Installing collected packages: pymysql
```

```
Successfully installed pymysql-1.1.1
```

[4]

```
import pymysql
```

[7]

```
import warnings
```

```
warnings.filterwarnings("ignore")
```

2.DATABASE CONNECTION:

[11]

```
connection=pymysql.connect(host='localhost',user='root',password='Swathi@12345',database='crime_data')
```

[12]

```
import pandas as pd
```

```
query= "select * from crime_data;"
```

[13]

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

[14]

```
df=pd.read_sql(query,connection)
```

```
print(df)
```

```
   DR_NO  Date_Rptd  DATE_OCC  AREA_NAME  Crm_Cd \
0  10304468  01-08-2020  01-08-2020  Southwest   624
1  190101086  01-02-2020  01-01-2020   Central   624
2  191501505  01-01-2020  01-01-2020  N Hollywood   745
3  191921269  01-01-2020  01-01-2020   Mission   740
4  200100502  01-02-2020  01-02-2020   Central   442
..     ...      ...      ...      ...      ...
494 200106614  02-07-2020  02-07-2020   Central   624
495 200106615  02-07-2020  02-07-2020   Central   624
```

```

496 200106616 02-07-2020 02-07-2020 Central 624
497 200106617 02-07-2020 01-10-2020 Central 510
498 200106618 02-07-2020 02-03-2020 Central 745

```

```

          Crm_Cd_Desc Vict_Age Vict_Sex \
0          BATTERY - SIMPLE ASSAULT    36    F
1          BATTERY - SIMPLE ASSAULT    25    M
2      VANDALISM - MISDEAMEANOR ($399 OR UNDER)    76    F
3  VANDALISM - FELONY ($400 & OVER, ALL CHURCH VA...    31    X
4      SHOPLIFTING - PETTY THEFT ($950 & UNDER)    23    M
..          ...      ...      ...
494          BATTERY - SIMPLE ASSAULT    33    M
495          BATTERY - SIMPLE ASSAULT    25    M
496          BATTERY - SIMPLE ASSAULT    35    M
497          VEHICLE - STOLEN         0    F
498      VANDALISM - MISDEAMEANOR ($399 OR UNDER)    72    M

```

```

          Premis_Desc Status      Location \
0      SINGLE FAMILY DWELLING  AO    1100 W 39TH PL
1          SIDEWALK    IC    700 S HILL ST
2  MULTI-UNIT DWELLING (APARTMENT, DUPLEX, ETC)  IC    5400 CORTEEN PL
3          BEAUTY SUPPLY STORE  IC    14400 TITUS ST
4          DEPARTMENT STORE    IC    700 S FIGUEROA ST
..          ...      ...      ...
494          SIDEWALK    IC    2400 ELLENDALE PL
495          SIDEWALK    IC          39TH PL
496          OTHER STORE    IC    2500 W VERNON AV
497          PARKING LOT    IC    700 EXPOSITION BL
498      VEHICLE, PASSENGER/TRUCK  IC    2600 S FIGUEROA ST

```

```

      LAT  LON
0  34.01 -118.30
1  34.05 -118.25
2  34.17 -118.40
3  34.22 -118.45
4  34.05 -118.26
..  ...  ...
494 34.05 -118.26
495 34.05 -118.26
496 34.05 -118.25
497 34.05 -118.25
498 34.05 -118.24

```

```
[499 rows x 13 columns]
```

```
[15]
```

```
print(type(df))
```

```
<class 'pandas.core.frame.DataFrame'>
```

3.DATA EXPLORATION:

```
[17]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 499 entries, 0 to 498
```

```
Data columns (total 13 columns):
```

```
#   Column      Non-Null Count  Dtype
---  -
0  DR_NO      499 non-null    int64
1  Date_Rptd  499 non-null    object
2  DATE_OCC   499 non-null    object
3  AREA_NAME  499 non-null    object
4  Crm_Cd     499 non-null    int64
5  Crm_Cd_Desc 499 non-null    object
6  Vict_Age   499 non-null    int64
7  Vict_Sex   499 non-null    object
8  Premis_Desc 499 non-null    object
9  Status     499 non-null    object
10 Location   499 non-null    object
11 LAT        499 non-null    float64
12 LON        499 non-null    float64
dtypes: float64(2), int64(3), object(8)
memory usage: 50.8+ KB
```

```
[18]
```

```
df.dtypes
```

```
DR_NO      int64
Date_Rptd  object
DATE_OCC   object
AREA_NAME  object
Crm_Cd     int64
Crm_Cd_Desc object
Vict_Age   int64
Vict_Sex   object
Premis_Desc object
Status     object
Location   object
LAT        float64
LON        float64
dtype: object
```

```
[19]
```

```
df.shape
```

```
(499, 13)
```

```
[20]
```

```
df.head()
```

```
[22]
```

```
df.tail()
```

```
[24]
```

```
df.describe()
```

```
[25]
```

```
df.describe(include='all')
```

```
[26]
```

```
df
```

```
unique_values=df["Crm_Cd"].unique()
```

```
print(unique_values)
```

```
[624 745 740 442 946 341 330 930 648 354 230 761 350 310 480 623 440 510  
210 900 888 420 886 421 647 940 662 220 625 755 649 901 320 890 351 956  
820 812 920 850 666 343]
```

```
[27]
```

```
df
```

```
unique_values=df["AREA_NAME"].unique()
```

```
print(unique_values)
```

```
['Southwest' 'Central' 'N Hollywood' 'Mission' 'Van Nuys' 'Wilshire'  
'Rampart' 'Harbor' 'Pacific' 'Hollenbeck' 'West LA' 'Northeast'  
'Foothill' 'Hollywood' 'West Valley' '77th Street' 'Olympic' 'Devonshire'  
'Southeast']
```

```
[28]
```

```
df
```

```
unique_values=df["Location"].unique()
```

```
print(unique_values)
```

```
['1100 W 39TH PL' '700 S HILL ST' '5400 CORTEEN PL' '14400 TITUS ST'  
'700 S FIGUEROA ST' '200 E 6TH ST' '700 BERNARD ST' '800 N ALAMEDA ST'  
'800 S OLIVE ST' '700 W 7TH ST' '100 S LOS ANGELES ST'  
'13600 LEADWELL ST' '8TH' 'WALL' '800 N BROADWAY' '400 S SPRING ST'  
'300 W 5TH ST' '1800 S MAIN ST' '600 W 7TH ST' '700 ALPINE ST'  
'6200 SYLMAR AV' '100 E 5TH ST' '1200 E 7TH ST' '700 S FLOWER ST'  
'300 E 3RD ST' '1ST' '1200 S GRAND AV' '300 S GRAND AV'  
'600 S CENTRAL AV' '600 N BROADWAY' '800 W 7TH ST' '7TH ST'  
'23400 PRESIDENT AV' '13100 LAKE ST' '100 W 1ST ST' '3RD' '6TH'  
'1600 S HOPE ST' '600 S SAN PEDRO ST' '100 W 2ND ST' '4TH']
```

'700 S GRAND AV' '5TH' 'PALMETTO' '800 JAMES M WOOD BL' 'GRAND'
'200 N AVENUE 25' '500 SAN JULIAN ST' '43RD ST' '100 PASEO DE LA PLAZA'
'500 S SAN PEDRO ST' '1300 S LOS ANGELES ST' '800 S FIGUEROA ST'
'JESSE ST' '100 E 4TH ST' '600 S SPRING ST' '7TH' '900 NEW DEPOT ST'
'400 CASANOVA ST' '600 S GRAND AV' '500 S SPRING ST' '600 WALL ST'
'900 N BROADWAY' '5TH ST' '700 S OLIVE ST' '100 N LOS ANGELES ST'
'6TH ST' '900 S LOS ANGELES ST' '200 N SOTO ST' 'HILL' '700 S BROADWAY'
'SPRING ST' '2200 SAN YSIDRO DR' '2000 E 7TH ST' 'W 5TH ST'
'800 S HILL ST' '100 S FIGUEROA ST' '600 S FIGUEROA ST' '100 E 17TH ST'
'6100 DELPHI ST' '400 S BROADWAY' '14TH' '300 S BROADWAY' '4TH ST' '12TH'
'8TH ST' '11TH ST' '300 E 5TH ST' '1ST ST' '200 W WASHINGTON BL'
'400 W 7TH ST' '1100 S FIGUEROA ST' '900 W 8TH ST' '400 SEATON ST'
'1000 N BROADWAY' '1300 MAGDALENA ST' '00 GATEWAY PLAZA DR'
'600 WILSHIRE BL' '500 N FIGUEROA ST' 'GEORGIA ST' 'ALPINE ST'
'500 WALL ST' 'PICO BL' '1100 S HOPE ST' '400 S HEWITT ST'
'1400 S FIGUEROA ST' 'VENICE' '200 S SAN PEDRO ST' '800 N HILL ST'
'700 S LOS ANGELES ST' '500 S BROADWAY' '400 SAVOY ST' '900 S HILL ST'
'W 17TH ST' '6500 YUCCA ST' '3900 DENKER AV' '700 N MAIN ST'
'700 WILSHIRE BL' '400 W 8TH ST' '1000 WILSHIRE BL' 'E 8TH ST'
'8700 SUNLAND BL' '300 E 2ND ST' '1300 S HOPE ST' '700 CENTER ST'
'500 S MAIN ST' 'FACTORY PL' '400 S ALAMEDA ST' '600 CERES AV' '2ND ST'
'WESTLAKE' 'HEWITT' '800 W OLYMPIC BL' '1800 S FLOWER ST' '700 S MAIN ST'
'OLIVE' '500 W 6TH ST' '700 N ALAMEDA ST' '600 E 5TH ST' '500 CERES AV'
'LOS ANGELES' '1600 SANTEE ST' '600 N FIGUEROA ST' '1000 S FLOWER ST'
'CESAR E CHAVEZ' '500 S HILL ST' '500 RAMIREZ ST' '900 N BEAUDRY AV'
'500 S LOS ANGELES ST' '600 N SPRING ST' '400 COLYTON ST' 'FLOWER'
'1200 S MAIN ST' '600 N ALAMEDA ST' '14600 VANOWEN ST'
'100 JUDGE JOHN AISO ST' '19500 INGOMAR ST' '200 W 7TH ST'
'600 S OLIVE ST' '900 W OLYMPIC BL' '200 BOYD ST' '200 E 11TH ST'
'1000 S HOPE ST' '1000 S FIGUEROA ST' 'FRIGATE' '1000 S ALVARADO ST'
'600 W 5TH ST' 'ORD' '200 N MAIN ST' 'SUNSET' '2600 S CABRILLO AV'
'100 S GRAND AV' '600 S LOS ANGELES ST' '800 W 255TH ST' '6300 6TH AV'
'7800 BEEMAN AV' '300 N VERMONT AV' '3300 SAN MARINO ST'
'15600 SIMONDS ST' '800 S TREMAINE AV' '2400 COLORADO BL' '3500 E 7TH ST'
'200 S WESTLAKE AV' '300 E 1ST ST' '400 HARTFORD AV' '1600 W 204TH ST'
'7900 BOTHWELL RD' '1500 VINE ST' '100 AVENUE 56' '900 S GRAND AV'
'6600 BRYNHURST AV' '1600 W 54TH ST' '6200 MT ANGELUS DR'
'200 WOODRUFF AV' '2200 MARICOPA DR' '3900 URSULA AV' '1400 W HILL DR'
'00 OLVERA ST' '600 W SUNSET BL' 'BOYD ST' '800 FRANCISCO ST' 'ALPINE'
'800 S GRAND AV' '700 S SPRING ST' '1200 S OLIVE ST' '400 WITMER ST'
'700 YALE ST' 'BROADWAY' '2ND' '100 W WASHINGTON BL' '900 S FIGUEROA ST'
'500 S FLOWER ST' '2200 W SUNSET BL' 'BURLINGTON' '1700 W 6TH ST'
'7900 WOODMAN AV' '2100 JAMES M WOOD BL' 'NORMAL AV' '400 LOMA DR'
'JAMES M WOOD' '1400 COURTLAND AV' '500 BRITANIA ST' 'CONNECTICUT'
'ADDISON' 'PICO' '600 S UNION AV' '2300 KENT ST' '3800 DUBLIN AV'
'SANTO TOMAS DR' '700 EXPOSITION BL' '3900 S WESTERN AV'
'3500 ENVILLE PL' '48TH ST' '1500 EXPOSITION BL' '1100 W 20TH ST'
'3400 S LA BREA AV' '900 W 41ST DR' '1500 W 37TH ST' '700 W 27TH ST'

'4000 S WESTERN AV' '3200 W VERNON AV' '1400 W 37TH PL' '1900 W 35TH PL'
 '9300 VAN NUYS BL' 'W 28TH ST' '3700 SANTA ROSALIA DR' '4000 STEVELY AV'
 '3600 CRENSHAW BL' '1200 W 30TH ST' '4000 CRENSHAW BL'
 '4100 S FIGUEROA ST' '4900 W ADAMS BL' '700 W WASHINGTON BL'
 '2700 S SYCAMORE AV' '2100 OAK ST' '800 W ADAMS BL' '10900 WELLWORTH AV'
 '3500 S FLOWER ST' '2400 S VERMONT AV' '2300 W 48TH ST'
 '4600 MARTIN LUTHER KING BL' '2000 W MARTIN LUTHER KING JR BL'
 '17800 MARGATE ST' '5000 OBAMA BL' 'ADAMS BL' '3900 S BUDLONG AV'
 '02900 S LA BREA AV' '2500 S FIGUEROA ST' '3600 WATT WY'
 '3600 S SAINT ANDREWS PL' '4300 BRIGHTON AV' '2400 W 48TH ST'
 '2700 S FIGUEROA ST' '4800 7TH AV' '4100 PALMWOOD DR' '600 W 42ND PL'
 'OBAMA BL' 'SANTA MONICA FY' '3500 S LA CIENEGA BL' '7TH AV'
 '2700 RAYMOND AV' '2100 W 29TH PL' '3400 EXPOSITION BL' '1100 W 39TH ST'
 '2500 CLYDE AV' '5400 GEER AV' '5500 CARLIN ST' '900 W JEFFERSON BL'
 '1100 W 28TH ST' '24100 GILMORE ST' '1200 W 37TH PL' '4200 S HARVARD BL'
 '1600 W 38TH ST' '2800 W 42ND ST' 'BRONSON AV' '5300 BLACKWELDER ST'
 '2600 RAYMOND AV' 'VERNON AV' '37TH ST' '1100 W 35TH ST' 'JEFFERSON BL'
 '3800 LOCKLAND DR' '4100 S HOOVER ST' '3000 ALSACE AV' '29TH'
 '1000 BAY VIEW AV' '4500 MARTIN LUTHER KING JR BL' '4000 WEST BL'
 '2300 W JEFFERSON BL' '1600 W 35TH ST' '700 W 30TH ST'
 '5000 EXPOSITION BL' '7300 SAUSALITO AV' '16100 VANOWEN ST'
 '400 N EVERGREEN AV' '13400 TERRA BELLA ST' '19100 JOVAN ST'
 '1900 SELBY AV' '100 W AVENUE 44' '5900 MONTEREY RD' '5600 W SUNSET BL'
 '1300 MIDVALE AV' 'EDGEMONT' '10700 WEYBURN AV' 'WILSHIRE'
 '9900 SATICOY ST' '5900 SPELTHORNE LN' 'CHEROKEE' '600 N SAN FERNANDO RD'
 '2700 S GRAND AV' '700 IMOGEN AV' 'ROSALIND PL' 'GRAND AV'
 '10300 COMPTON AV' '1300 W C ST' '2900 W OLYMPIC BL'
 '10900 SANTA MONICA BL' '3100 S FIGUEROA ST' '4000 ABOURNE RD'
 '1000 W 34TH ST' '126TH ST' '400 S SOTO ST' '700 STATE DR'
 '3700 CARMONA AV' '5100 OBAMA BL' '4300 3RD AV' '4700 7TH AV'
 '3600 W MARTIN LUTHER KING JR BL' '1600 W MARTIN LUTHER KING BL'
 '4300 S FIGUEROA ST' 'BUCKINGHAM' 'KENILWORTH AV' '3900 S VAN NESS AV'
 '1100 HARTZELL ST' '4100 S BUDLONG AV' 'FIGUEROA ST' '1200 W 39TH ST'
 '4200 S WESTERN AV' '4000 SANTO TOMAS DR' '600 EXPOSITION BL'
 '1400 W 37TH DR' '2400 ELLENDALE PL' '39TH PL' '2500 W VERNON AV'
 '2600 S FIGUEROA ST']

[32]

df

```
unique_values=df["Crm_Cd_Desc"].unique()
```

```
print(unique_values)
```

['BATTERY - SIMPLE ASSAULT' 'VANDALISM - MISDEAMEANOR (\$399 OR UNDER)'

'VANDALISM - FELONY (\$400 & OVER, ALL CHURCH VANDALISMS)'

'SHOPLIFTING - PETTY THEFT (\$950 & UNDER)' 'OTHER MISCELLANEOUS CRIME'

'THEFT-GRAND (\$950.01 & OVER)EXCPT,GUNS,FOWL,LIVESTK,PROD'

'BURGLARY FROM VEHICLE' 'CRIMINAL THREATS - NO WEAPON DISPLAYED' 'ARSON'

'THEFT OF IDENTITY' 'ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT'

```
'BRANDISH WEAPON' 'THEFT, PERSON' 'BURGLARY' 'BIKE - STOLEN'
'BATTERY POLICE (SIMPLE)' 'THEFT PLAIN - PETTY ($950 & UNDER)'
'VEHICLE - STOLEN' 'ROBBERY' 'VIOLATION OF COURT ORDER' 'TRESPASSING'
'THEFT FROM MOTOR VEHICLE - PETTY ($950 & UNDER)' 'DISTURBING THE PEACE'
'THEFT FROM MOTOR VEHICLE - ATTEMPT' 'THROWING OBJECT AT MOVING VEHICLE'
'EXTORTION' 'BUNCO, GRAND THEFT' 'ATTEMPTED ROBBERY' 'OTHER ASSAULT'
'BOMB SCARE' 'DOCUMENT FORGERY / STOLEN FELONY'
'VIOLATION OF RESTRAINING ORDER' 'BURGLARY, ATTEMPTED' 'FAILURE TO YIELD'
'PURSE SNATCHING' 'LETTERS, LEWD - TELEPHONE CALLS, LEWD'
'ORAL COPULATION'
'CRM AGNST CHLD (13 OR UNDER) (14-15 & SUSP 10 YRS OLDER)'
'KIDNAPPING - GRAND ATTEMPT' 'INDECENT EXPOSURE' 'BUNCO, ATTEMPT'
'SHOPLIFTING-GRAND THEFT ($950.01 & OVER)']
```

4.TEMPORAL ANALYSIS

CRIME OCCURANCE OVER TIME:

[34]

```
df['Date_Rptd']=pd.to_datetime(df['Date_Rptd'])
df['DATE_OCC']=pd.to_datetime(df['DATE_OCC'])
```

[35]

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 499 entries, 0 to 498
```

```
Data columns (total 13 columns):
```

```
#   Column      Non-Null Count  Dtype
```

```
---  ---
```

```
0  DR_NO      499 non-null    int64
1  Date_Rptd  499 non-null    datetime64[ns]
2  DATE_OCC   499 non-null    datetime64[ns]
3  AREA_NAME  499 non-null    object
4  Crm_Cd     499 non-null    int64
5  Crm_Cd_Desc 499 non-null    object
6  Vict_Age   499 non-null    int64
7  Vict_Sex   499 non-null    object
8  Premis_Desc 499 non-null    object
9  Status     499 non-null    object
10 Location   499 non-null    object
11 LAT        499 non-null    float64
12 LON        499 non-null    float64
```

```
dtypes: datetime64[ns](2), float64(2), int64(3), object(6)
```

```
memory usage: 50.8+ KB
```

5.SPATIAL ANALYSIS

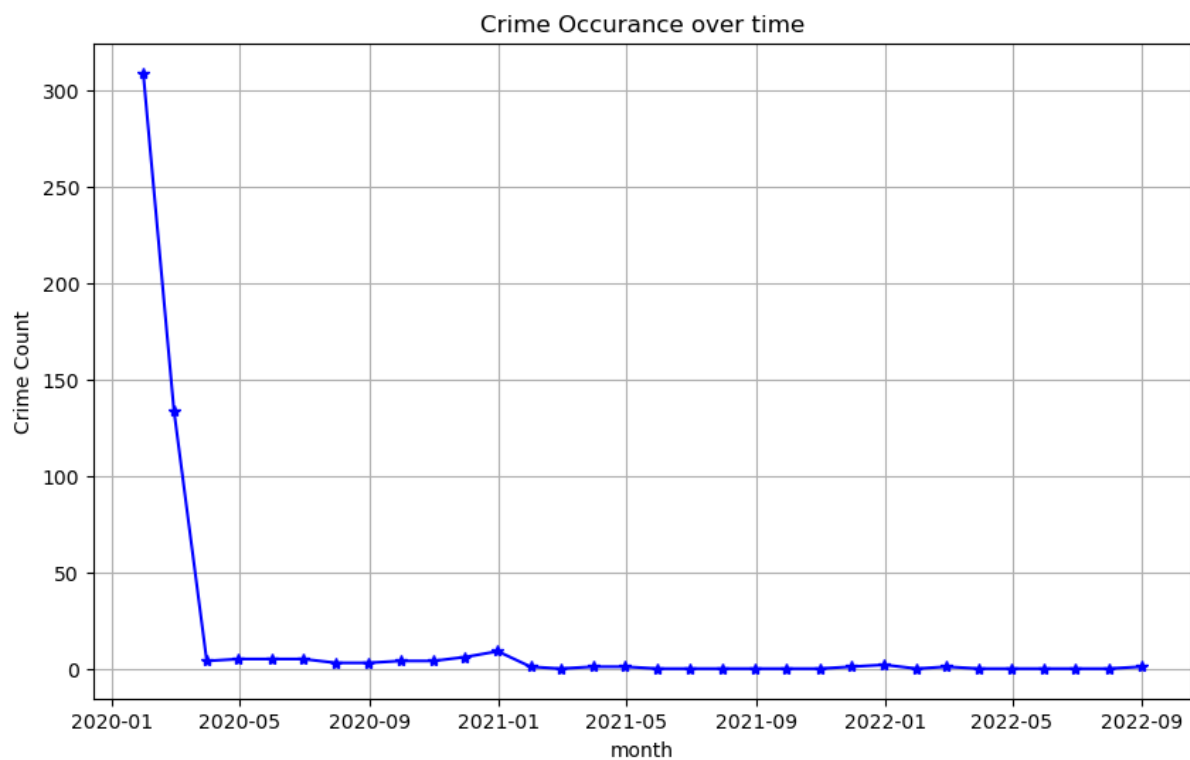
#Utilize the geographical information (Latitude and Longitude) to perform spatial analysis.

[40]

```
m=df.resample('M',on='Date_Rptd').size()
```

```
[41]
```

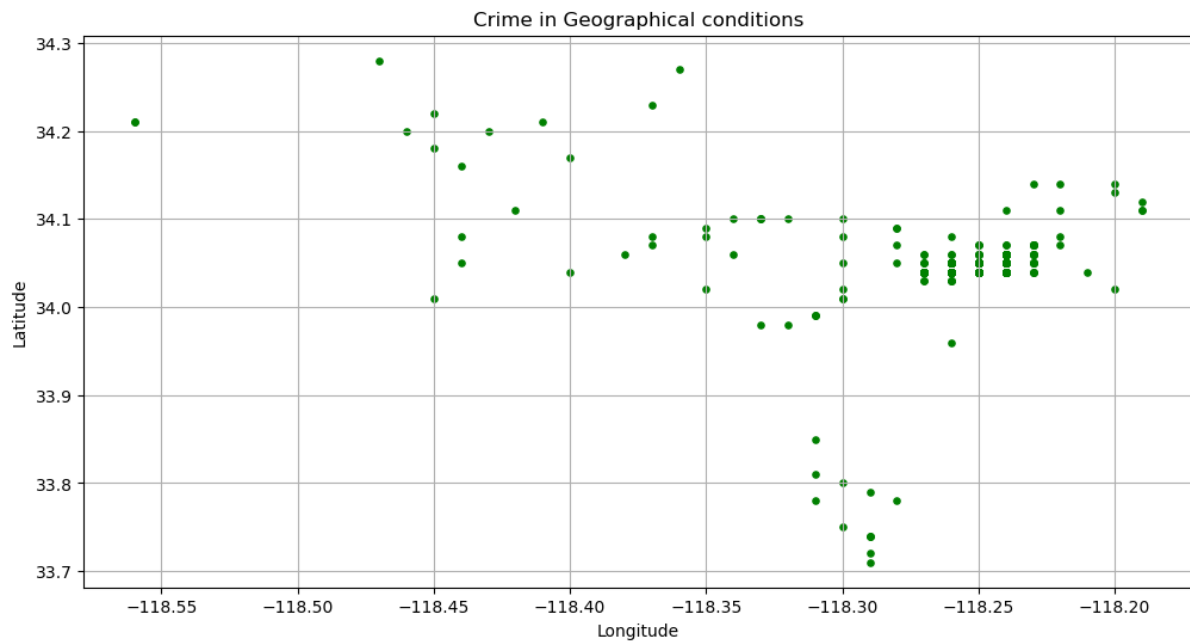
```
plt.figure(figsize=(10,6))
plt.plot(m.index,m.values,marker='*',color='blue')
plt.title('Crime Occurance over time')
plt.xlabel('month')
plt.ylabel('Crime Count')
plt.grid()
plt.show()
```



```
# Visualize crime hotspots on a map.
```

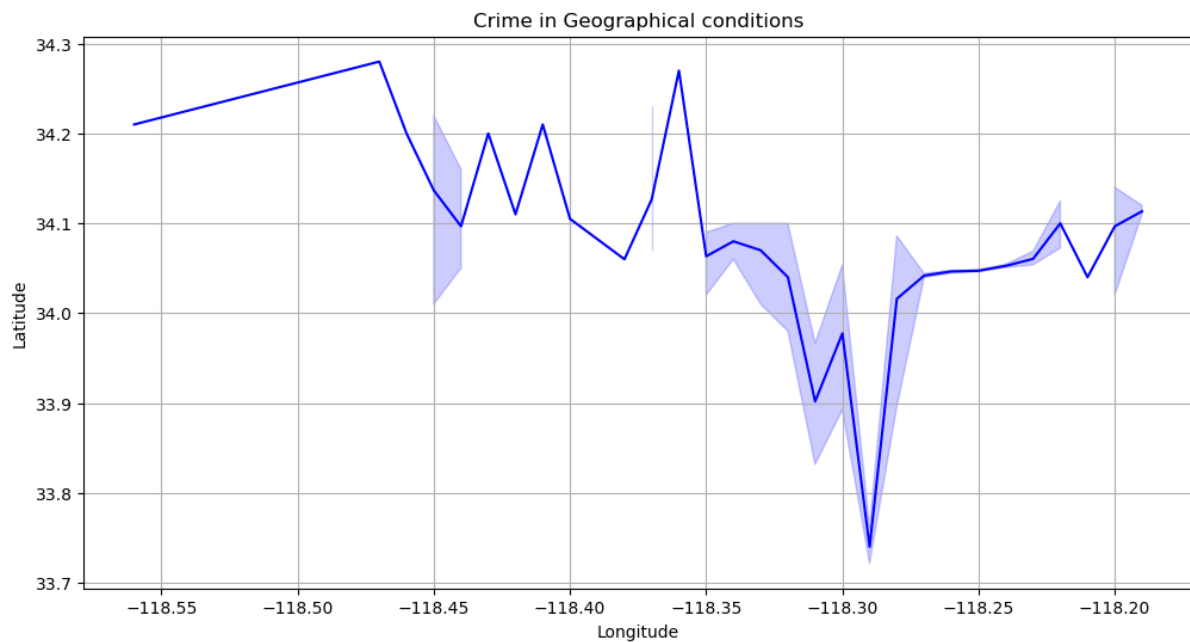
```
[52]
```

```
plt.figure(figsize=(12,6))
plt.scatter(df['LON'],df['LAT'],s=14,color='green')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title('Crime in Geographical conditions')
plt.grid()
plt.show()
```



[55]

```
plt.figure(figsize=(12,6))
sns.lineplot(x=df['LON'],y=df['LAT'],color='blue')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title('Crime in Geographical conditions')
plt.grid()
plt.show()
```



[56]

```
location_count=df['Location'].value_counts()
print(location_count)
```

```

Location
800 N ALAMEDA ST    14
700 W 7TH ST        11
1100 S FIGUEROA ST   6
300 E 5TH ST         6
6TH                  6
..
14600 VANOWEN ST     1
600 N ALAMEDA ST     1
FLOWER               1
400 COLYTON ST       1
2600 S FIGUEROA ST   1
Name: count, Length: 365, dtype: int64

```

```
[59]
```

```

#based on crime code
crime_code=df['Crm_Cd']
crime_count=crime_code.value_counts()
print(crime_count)

```

```

Crm_Cd
330    82
624    74
440    44
442    28
510    28
341    26
230    24
740    23
745    23
888    19
310    16
210    12
480    11
354    10
420     9
350     9
930     7
623     5
761     5
662     5
946     4
220     4
648     3
901     3
850     2
956     2
351     2

```

```

320  2
625  2
886  2
900  2
755  1
649  1
940  1
647  1
890  1
421  1
820  1
812  1
920  1
666  1
343  1
Name: count, dtype: int64

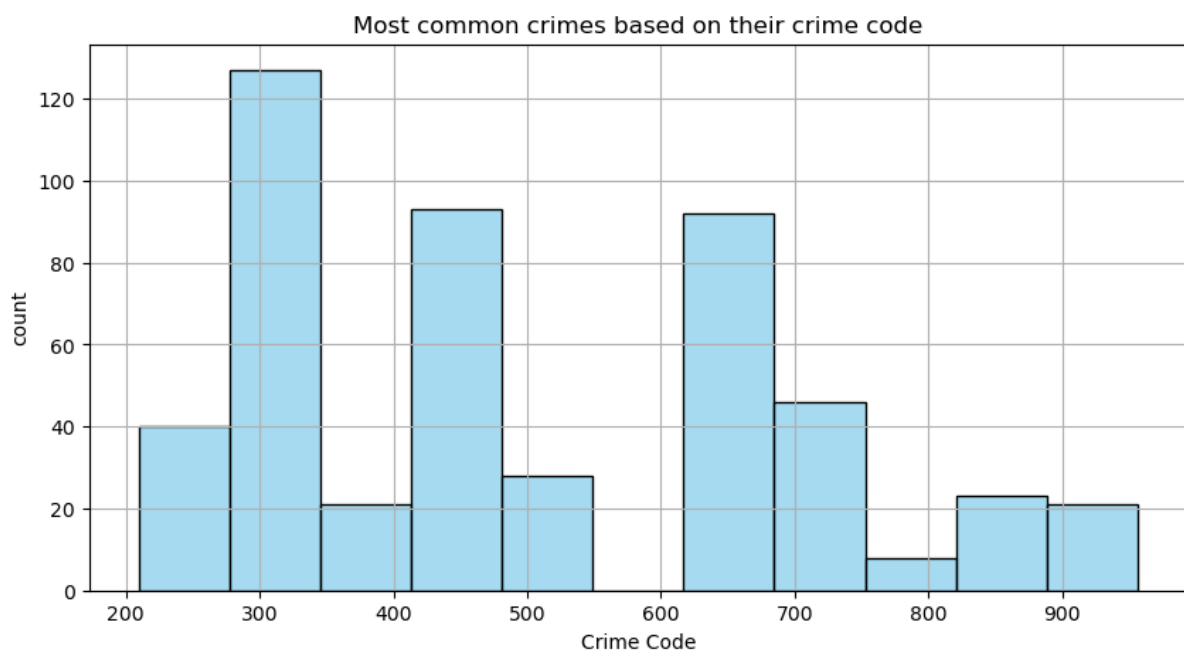
```

```
[62]
```

```

plt.figure(figsize=(10,5))
sns.histplot(df['Crm_Cd'],color='skyblue',edgecolor='black')
plt.title('Most common crimes based on their crime code')
plt.xlabel('Crime Code')
plt.ylabel('count')
plt.grid()
plt.show()

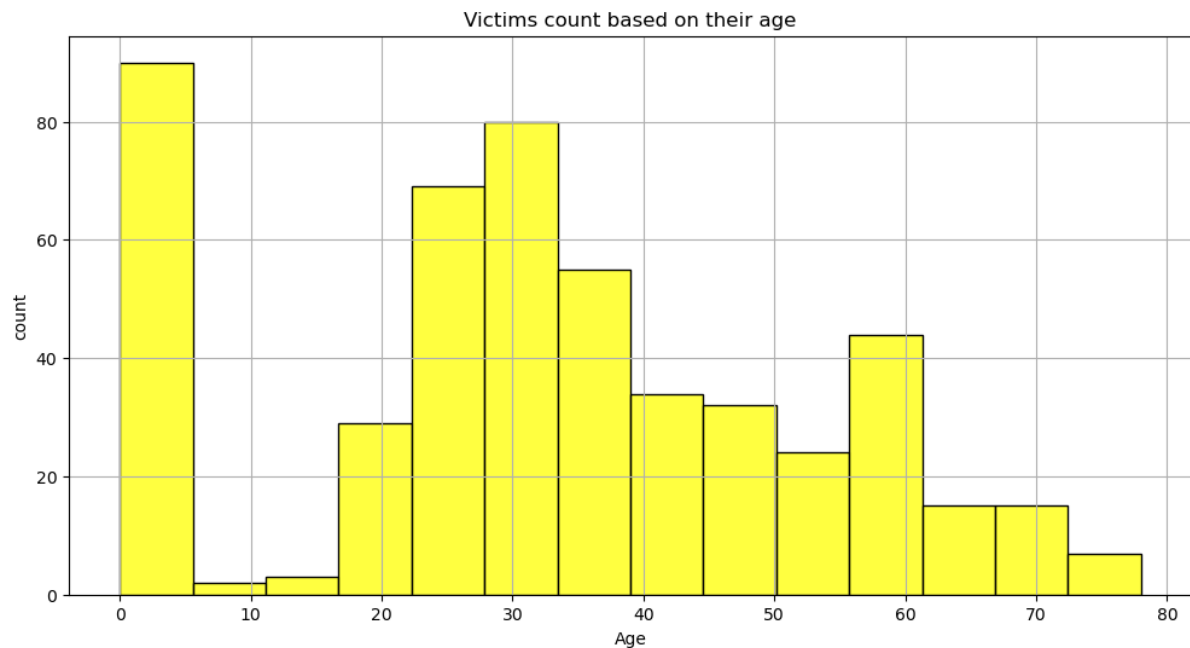
```



6.Victim Demographics:
#Investigate the distribution of victim ages and genders

```
[64]
```

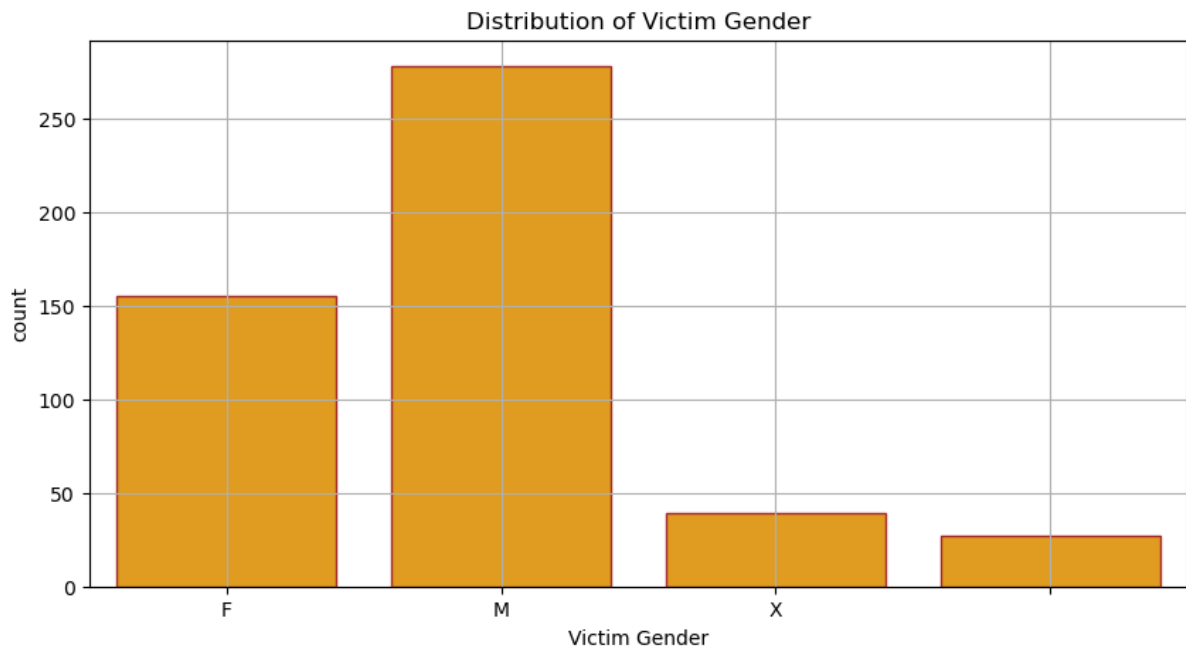
```
plt.figure(figsize=(12,6))
sns.histplot(df['Vict_Age'],color='yellow',edgecolor='black')
plt.title('Victims count based on their age')
plt.xlabel('Age')
plt.ylabel('count')
plt.grid()
plt.show()
```



#here 0 indicates to the people whose age is not mentioned or not defined in the given data.

[70]

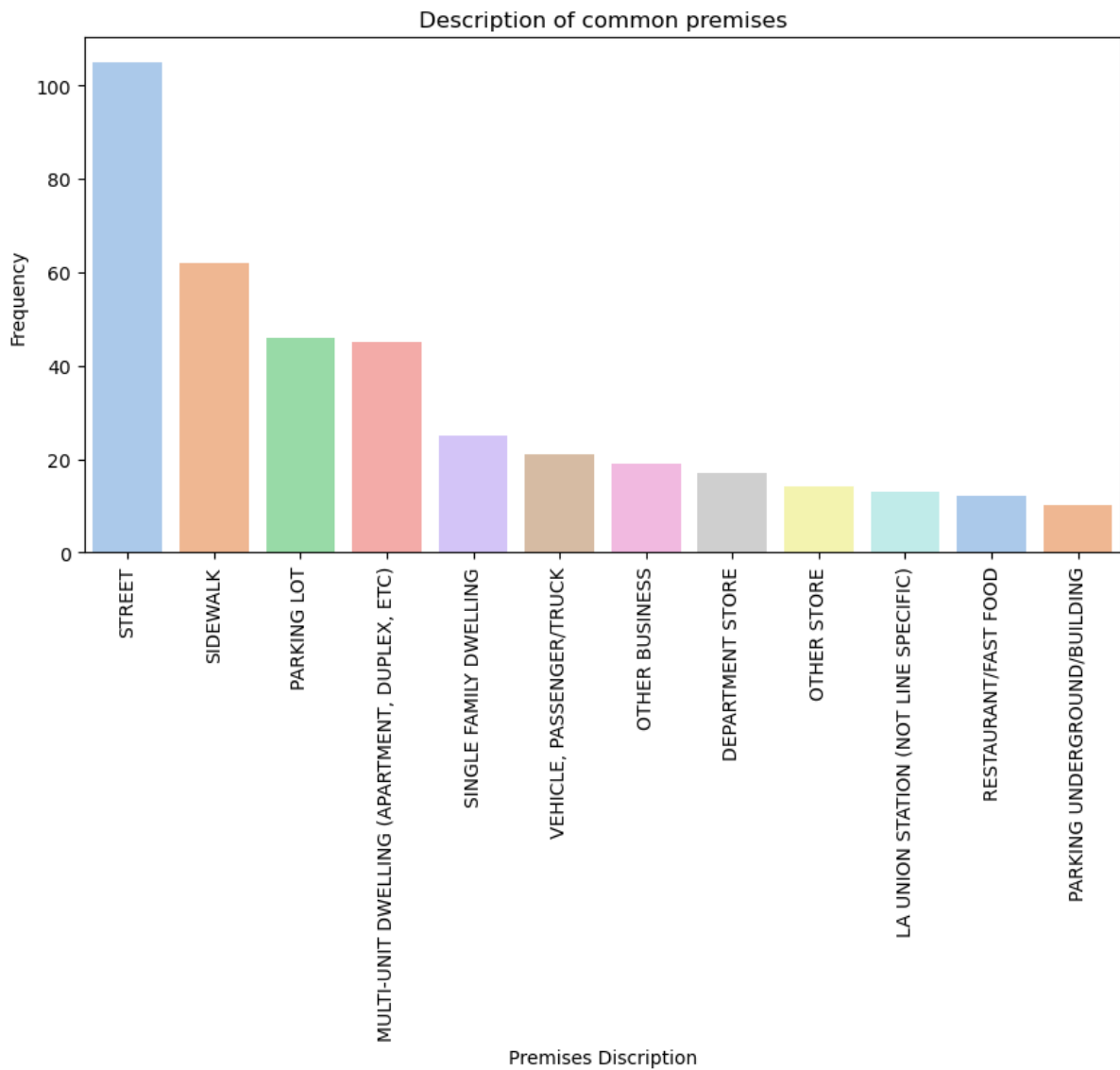
```
plt.figure(figsize=(10,5))
sns.countplot(x=df['Vict_Sex'],edgecolor='brown',color='orange')
plt.xlabel('Victim Gender')
plt.title('Distribution of Victim Gender')
plt.grid()
plt.show()
```



Identify common premises descriptions where crimes occur.

[75]

```
plt.figure(figsize=(10,5))
sns.countplot(x=df['Premis_Desc'],order=df['Premis_Desc'].value_counts().index[:12],palette='pastel'
)
plt.xlabel('Premises Discription')
plt.ylabel('Frequency')
plt.title('Description of common premises')
plt.xticks(rotation=90)
plt.show()
```



[77]

```
Q=df['Premis_Desc']
count=Q.value_counts()
print(count)
```

```
Premis_Desc
STREET                105
SIDEWALK              62
PARKING LOT           46
MULTI-UNIT DWELLING (APARTMENT, DUPLEX, ETC)  45
SINGLE FAMILY DWELLING  25
...
MINI-MART              1
SINGLE RESIDENCE OCCUPANCY (SRO'S) LOCATIONS  1
MEDICAL/DENTAL OFFICES  1
DETENTION/JAIL FACILITY  1
```

MTA PROPERTY OR PARKING LOT 1

Name: count, Length: 67, dtype: int64

7.STATUS ANALYSIS:

#Examine the status of reported crimes.

[78]

```
W=df['Status']
status_count=W.value_counts()
print(status_count)
```

Status

IC 447

AA 27

AO 24

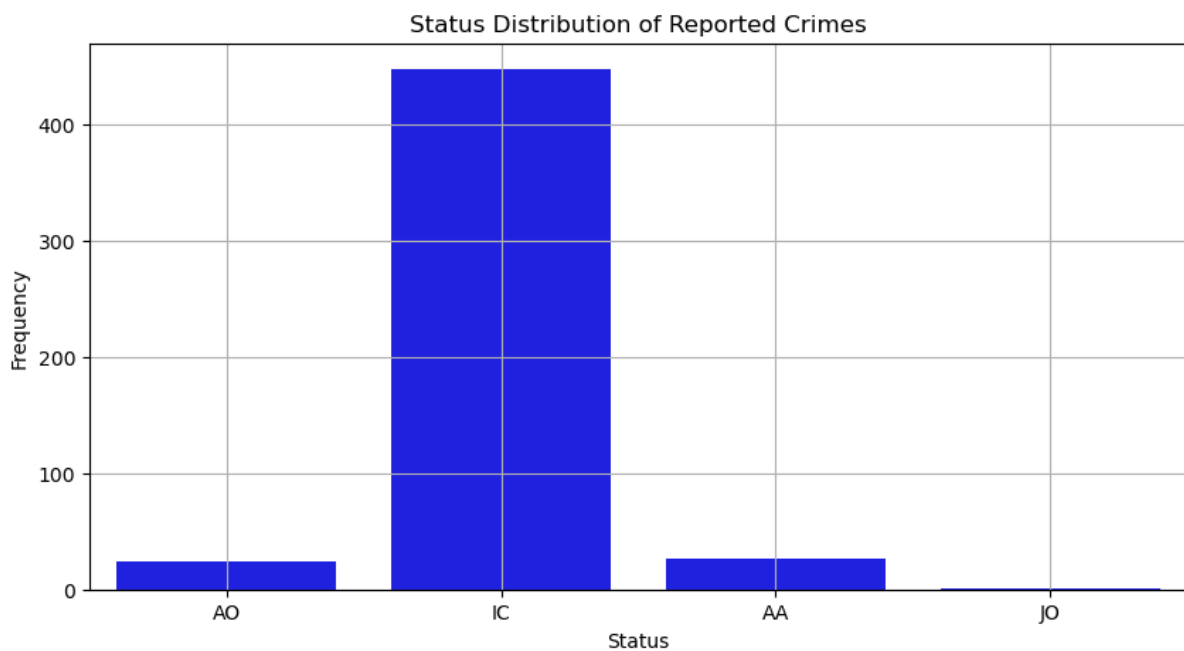
JO 1

Name: count, dtype: int64

#Classify crimes based on their current status.

[80]

```
plt.figure(figsize=(10,5))
sns.countplot(x=df['Status'],color='blue')
plt.xlabel('Status')
plt.ylabel('Frequency')
plt.title('Status Distribution of Reported Crimes')
plt.grid()
plt.show()
```



Questions:

Spatial Analysis:

1.Where are the geographical hotspots **for** reported crimes?

A. The Geographical hotspots **for** reported crimes **is** between the latitude 34.0 to 34.1 **and** longitude -118.30 to -118.20.

Victim Demographics:

2.What **is** the distribution of victim ages **in** reported crimes?

A. The age group "0" are higher.Apart **from** that the age group between 25 to 34 reported high crime rate.Least reported crime age group are below 15 age **and** 75 to 80 age group.

3.Is there a significant difference **in** crime rates between male **and** female victims?

A. Yes, there a significant difference **in** crime rates between male **and** female victims.The male crime rate **is** above 250 **and** the female crime rate **is** above 150.

Location Analysis:

4. Where do most crimes occur based on the Location column?

A. Based on the Location column the most crimes occur **in** ALAMEDA ST(14) followed by 7th ST(11).

[81]

```
location_count=df['Location'].value_counts()
print(location_count)
```

Location

```
800 N ALAMEDA ST    14
700 W 7TH ST        11
1100 S FIGUEROA ST   6
300 E 5TH ST         6
6TH                  6
```

..

```
14600 VANOWEN ST    1
600 N ALAMEDA ST    1
FLOWER              1
400 COLYTON ST      1
2600 S FIGUEROA ST  1
```

Name: count, Length: 365, dtype: int64

Crime Code Analysis:

5.What **is** the distribution of reported crimes based on Crime Code?

A. The Crime code 330 reported highest crimes of count 82 followed by crime code 624,440,442 **and** so on.

Below are the top 5 reported crimes based on crime code.

[85]

```
crime_code=df['Crm_Cd']  
crime_count=crime_code.value_counts()  
print(crime_count.head())
```

Crm_Cd

330 82

624 74

440 44

442 28

510 28

Name: count, dtype: int64